

Last name _____

First name _____

LARSON—MATH 310—CLASSROOM WORKSHEET 01
Vectors.

1. If $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$ and $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, what is the vector $\vec{w} = \begin{bmatrix} \vec{v} \\ \vec{u} \end{bmatrix}$

2. What does $\mathbf{nnz}(\vec{v})$ mean?

3. If $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$, what is $\mathbf{nnz}(\vec{v})$?

4. If $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$ and $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, what is $\vec{v} + \vec{u}$?

5. If $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$ and $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, what is $\vec{v} - \vec{u}$?

6. If $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$, what is $4\vec{v}$?

7. What is a *linear combination*?

8. If $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$ and $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, find the linear combination $4\vec{v} - 3\vec{u}$.